

Prime-Indexed Recursive Governance: A Multiplicity-Theoretic Formalization of the Competence Protocol

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Citizen Gardens
The Foundation of Multiplicity

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Abstract

This defensive publication presents a complete, citable description of a novel governance framework that integrates Multiplicity Theory—specifically Prime-Indexed Recursive Tensor Mathematics (PIRTM)—with a practical political selection and evaluation system called the *Competence Protocol*. The core invention is a prime-indexed, recursively updated evolution operator acting on a Hilbert space of governance states, driven by publicly observable reasoning (*type-talk*), long-term consequence prediction, and real-world outcome feedback. The publication defines (1) a canonical mapping from 10 governance competence pillars to prime-labeled sectors, (2) a discrete–continuous evolution law $X_{t+1} = \Xi(t)X_t + \Lambda_m(t)T(X_t)$ for collective decision states, (3) stability conditions guaranteeing convergence of governance trajectories toward a bounded, competence-dominated attractor, (4) spectral drift metrics that trigger automatic public cross-examination, and (5) an implementable protocol for municipal and national pilots including live broadcast, real-time scoring, and open-source analytical tooling. This document is intended as both a scientific report and a prior-art defensive publication, preventing exclusive patenting of prime-indexed, multiplicity-based governance selection and evaluation systems.

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1 Executive Summary

1.1 Invention Overview

This publication formalizes a new class of governance systems in which political leaders and decision-making bodies are selected, evaluated, and continuously updated via a mathematically rigorous, prime-indexed, recursive framework.

The core elements are:

- A **Competence Protocol**: a 5–7 day live, humane assessment of leadership candidates, including deep testing across 10 competence pillars, real-time reasoning (*type-talk*), and 5/10/20-year consequence mapping for policies.
- A **Prime-Indexed Embedding**: mapping each pillar to a distinct prime-labeled sector $p \in \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29\}$, yielding a structured multiplicity space for competence.
- A **Governance Evolution Operator**: an operator $\Xi(t)$ constructed from live reasoning and feedback, plus a multiplicity scalar $\Lambda_m(t)$ updated by outcome coherence, generating the evolution

$$X_{t+1} = \Xi(t)X_t + \Lambda_m(t)T(X_t),$$

where X_t is the state of governance at time t .

- **Stability and Contraction**: conditions under which trajectories of governance states converge to a unique bounded attractor dominated by competent leaders, while incompetent trajectories decay.
- **Spectral Drift and Meta-Governance**: a drift metric detects decoherence in leader reasoning and triggers public cross-examination, while the assessment framework itself is recursively updated.

The system transforms politics into a measurable, contractive, self-stabilizing process governed by multiplicity-theoretic operators rather than ad hoc charisma-based selection.

1.2 Problem Addressed

Existing political systems typically select leaders via popularity mechanisms (elections) that:

- reward charisma over demonstrated competence;
- provide limited, opaque feedback loops from long-term policy outcomes;
- make reasoning and trade-offs largely invisible to the public;
- lack formally defined, recursively updated competence metrics.

The invention addresses these deficiencies by:

1. making reasoning and trade-offs publicly observable via type-talk and live broadcast;
2. requiring explicit long-term consequence mapping with later outcome validation;
3. embedding leader competence in a prime-indexed multiplicity framework that admits formal stability analysis;
4. enabling continuous evaluation and reallocation of power according to mathematically defined reputation signatures.

1.3 Core Contributions

This defensive publication discloses, in a manner enabling skilled implementation:

1. A prime-indexed mapping of governance competence pillars to operators on a Hilbert space of decision states.
2. A discrete–continuous evolution law for governance with explicit stability conditions.
3. A formal definition of coherence, drift, and multiplicity-based reputation.
4. A practical protocol for real-world pilots (e.g., municipal-level trials), including broadcast, scoring, and open-source tooling.
5. Example pseudocode and data-flow implementation patterns.

This combination, with the specific structure described below, is asserted as prior art.

2 Conceptual Overview

2.1 Multiplicity-Theoretic Background

Multiplicity Theory interprets mathematical and physical systems as recursively generated patterns of prime-labeled interactions, rather than as static objects. In this framework:

- Primes index fundamental sectors of behavior or capacity.
- Objects (including governance states) are represented as elements in a multiplicity space, often modeled as a Hilbert space with prime-graded projectors.
- Dynamics are described via evolution operators $\Xi(t)$ and multiplicity scalars $\Lambda_m(t)$ that modulate nonlinear transforms T .

The Universal Multiplicity Constant Λ_m is treated as a governing scalar controlling the strength of recursive feedback between predictions and outcomes across scales.

2.2 From Competence Protocol to Evolution Operator

The original Competence Protocol specifies:

- intensive 5–7 day testing of candidates;
- 10 pillars of competence (problem-solving, integrity, economic understanding, etc.);
- live broadcast and type-talk reasoning;
- long-term consequence forecasting and later validation;
- annual or periodic refresh tests for those in office.

By embedding this protocol into the Multiplicity Theoretic framework, we interpret:

- each pillar as a prime-labeled sector p ,

- each candidate or team as a vector X_t in a Hilbert space $\mathcal{H}$,
- each decision cycle as an application of a time-varying operator $\Xi(t)$ plus a multiplicity-scaled nonlinear transform T .

The resulting system is both:

1. a rigorous mathematical evolution on $\mathcal{H}$, and
2. a practically implementable governance mechanism.

3 Prime-Sector Mapping of Governance Competence

3.1 Prime Set and Pillar Assignment

We define the prime set:

$$P_{10} = \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29\}.$$

Each prime $p \in P_{10}$ corresponds to a specific governance competence pillar:

Prime p	Pillar Domain	Informal Description
2	Problem-Solving Simulations	Crisis response under live scenarios
3	Long-Term Consequence Mapping	5/10/20-year impact prediction fidelity
5	Integrity & Anti-Corruption	Ethical drift resistance under pressure
7	Team/Ministerial Coherence	Quality of collective decision processes
11	Economic & Resource Allocation	Quantitative policy and trade-off literacy
13	Social Unrest & Justice Dynamics	Handling rights, welfare, conflict
17	Innovation & Adaptive Systems	Handling rapidly changing environments
19	Transparency & Type-Talk	Clarity and honesty of public reasoning
23	Cross-Scale Fractal Reasoning	Consistency from local to global scale
29	Meta-Governance Self-Recursion	Ability to improve the system itself

3.2 Sectoral Operators and Weights

For each time step t (e.g., a decision cycle or assessment segment) and prime $p \in P_{10}$, we define:

- $U_p(t)$: a sectoral operator on $\mathcal{H}$, derived from:
 - parsed type-talk transcripts,
 - task performance in simulations,
 - expert scoring of reasoning quality.
- $w_p(t) \in \mathbb{R}$: a real-valued weight derived from:
 - real-time expert scores,
 - citizen feedback and Q&A,
 - integrity and consistency checks.

The time-varying evolution operator is:

$$\Xi(t) = \sum_{p \in P_{10}} w_p(t) U_p(t).$$

We require a contractivity condition:

$$\|\Xi(t)\| \leq 1 - \varepsilon_*,$$

for some $\varepsilon_* > 0$ and all t . This guarantees that, in the absence of the multiplicity-driven nonlinear term, the system is uniformly stable.

4 Governance State Space and Evolution Law

4.1 Governance Hilbert Space

Let $\mathcal{H}$ be a (real or complex) Hilbert space representing governance states. Each state $X_t \in \mathcal{H}$ at time t captures:

- the aggregated competence profiles of leaders in office;
- the active policy configuration;
- encoded public trust and legitimacy dimensions (if desired as additional components).

Candidates or teams can be represented as vectors in a subspace of $\mathcal{H}$, while the full national state may be a higher-dimensional composite.

4.2 Nonlinear Transform and Coherence Function

We define a nonlinear transform $T : \mathcal{H} \rightarrow \mathcal{H}$ that encodes prediction–outcome error. For each decision D with predicted and actual impacts at time-offsets 5, 10, 20 years, we define a scalar coherence function:

$$\Gamma(D, t) = \sum_{k \in \{5, 10, 20\}} w_k \frac{|A_k - P_k|}{\max(A_k, P_k)},$$

where:

- P_k : predicted impact at horizon k ,
- A_k : actual observed impact at horizon k ,
- w_k : temporal weights (possibly prime-weighted).

The transform $T(X_t)$ aggregates these coherence values over past decisions encoded in X_t , e.g., by:

- projecting X_t onto decision subspaces,
- scaling those components by functions of Γ ,
- recombining them to produce a feedback vector.

We assume T is Lipschitz continuous with constant L_T :

$$\|T(X) - T(Y)\| \leq L_T \|X - Y\|, \quad \forall X, Y \in \mathcal{H}.$$

4.3 Multiplicity Scalar $\Lambda_m(t)$

The multiplicity scalar $\Lambda_m(t) \in \mathbb{R}$ modulates the feedback from T . It is updated based on coherence:

$$\Lambda_m(t+1) = F_\Lambda(\Lambda_m(t), \{\Gamma(D, t)\}_D),$$

for some update rule F_Λ , such as:

$$\Lambda_m(t+1) = \Lambda_m(t) \cdot \exp(-\alpha \bar{\Gamma}(t)),$$

where $\bar{\Gamma}(t)$ is the aggregated coherence across decisions at time t , and $\alpha > 0$ is a sensitivity parameter.

We define the multiplicity coupling constant:

$$c = \sup_t \|\Lambda_m(t)\| L_T.$$

4.4 Discrete-Time Evolution Law

The discrete-time evolution equation for governance state is:

$$X_{t+1} = \Xi(t)X_t + \Lambda_m(t)T(X_t).$$

Interpretation:

- $\Xi(t)$ propagates the state based on the currently observed reasoning and structural decision-making;
- $\Lambda_m(t)T(X_t)$ adjusts the state based on long-term prediction accuracy and outcome coherence.

4.5 Continuous-Time Evolution Law

For continuous live governance streams, we write:

$$\dot{X}(\tau) = F(\tau)X(\tau) + \Lambda_m(\tau)T(X(\tau)),$$

where $F(\tau)$ is the strong-operator derivative of Ξ ; i.e., $\Xi(\tau + d\tau) \approx \Xi(\tau) + F(\tau)d\tau$.

5 Stability, Contraction, and Attractors

5.1 Stability Theorem

Theorem (Stability and Contraction). Suppose:

1. $\|\Xi(t)\| \leq 1 - \varepsilon$ for some $\varepsilon > 0$ and all t .
2. T is Lipschitz with constant L_T .
3. Multiplicity coupling satisfies $c = \sup_t \|\Lambda_m(t)\| L_T < \varepsilon$.

Then for any two initial states $X_0, Y_0 \in \mathcal{H}$, the resulting trajectories satisfy:

$$\|X_t - Y_t\| \leq (1 - \varepsilon + c)^t \|X_0 - Y_0\|,$$

and hence:

$$\|X_t - Y_t\| \rightarrow 0 \quad \text{as } t \rightarrow \infty.$$

Interpretation. Under these conditions, the evolution is a contraction mapping on average, and governance trajectories converge toward a unique bounded attractor, dominated by competence profiles that minimize prediction error (Γ).

5.2 Governance Implications

- Competent leaders with accurate prediction and coherent reasoning are reinforced via $\Lambda_m(t)$, driving X_t toward their associated subspace.
- Incompetent or incoherent trajectories are exponentially suppressed and driven toward a “spectral zero” subspace (negligible influence).
- The system is robust to initial conditions; even legacy, charisma-driven states converge under sustained application of the protocol.

6 Spectral Drift, Cross-Examination, and Meta-Governance

6.1 Spectral Drift Metric

Let $\Phi(p, t)$ be a phase-like observable for sector p , derived from type-talk and performance in that sector. We define spectral drift:

$$\delta_{\text{drift}}(t) = \sum_{p \in P_{10}} \kappa_p [\Phi(p, t) - \Phi(p, t-1)],$$

with coefficients κ_p encoding sectoral importance.

If $|\delta_{\text{drift}}(t)|$ exceeds a threshold, the system triggers:

- automatic cross-examination of the relevant leader(s),
- live public questioning,
- or initiation of recall/replacement processes as specified by governance rules.

6.2 Spectral Gates

We may define a spectral projector decomposition:

$$I = \sum_{\alpha} E_{\alpha}(t),$$

where $E_{\alpha}(t)$ projects onto specific bounded spectral bands. After each evolution step, we optionally apply:

$$X_{t+1} \leftarrow \sum_{\alpha} E_{\alpha}(t) X_{t+1},$$

enforcing additional structural constraints (e.g., rule-of-law bounds, constitutional limits).

6.3 Meta-Governance Self-Recursion

The 10-pillar framework itself is subject to recursive evaluation. Define:

$$\eta = \frac{\text{actual overall governance quality}}{\text{predicted quality given high-scoring leaders}},$$

where governance quality can be measured by composite indices (e.g., public trust, economic stability, conflict levels). If η is systematically below 1, we adjust:

- the weights of pillars,
- the design of tasks associated with each prime sector,
- or the set P_{10} itself (meta-innovation).

Thus, not only leaders but also the *system design* is continuously corrected by the same multiplicity-based logic.

7 Reputation Ledger and Prime Multiplicity Signature

7.1 Prime-Indexed Competence Signature

For each leader or team, we define a prime-indexed signature:

$$R = \prod_{i=1}^{10} p_i^{s_i},$$

where $p_i \in P_{10}$ and $s_i \in \mathbb{R}$ encodes demonstrated strength in sector i , aggregated over time from:

- scoring during assessments,
- long-term coherence of predictions,
- observed behavior during governance.

7.2 Emergent Role Allocation

Leadership roles are allocated dynamically by:

- computing domain-specific signatures (e.g., economics-focused roles use exponents for $p = 11, 17, 23$ more heavily);
- assigning responsibilities to candidates with highest relevant R -values;
- continually updating these allocations as new data arrives.

This replaces static office tenure with fluid, competence-weighted responsibility assignments.

8 Implementation Blueprint and Pseudocode

8.1 High-Level Data Flow

1. Input Streams:

- Live type-talk transcripts (spoken or typed reasoning).
- Simulation performance metrics.
- Expert and citizen scoring.
- Real-world outcomes for previously decided policies.

2. Processing:

- Parse transcripts to extract sectoral operators $U_p(t)$.
- Compute sectoral weights $w_p(t)$.
- Build $\Xi(t) = \sum_p w_p(t)U_p(t)$.
- Evaluate coherence $\Gamma(D, t)$ for each tracked decision.
- Update $\Lambda_m(t)$ from coherence.

3. State Update:

- Apply $X_{t+1} = \Xi(t)X_t + \Lambda_m(t)T(X_t)$.
- Optionally apply spectral gates.
- Compute $\delta_{\text{drift}}(t)$ and respond if thresholds are crossed.
- Update reputation ledger R .

8.2 Example Pseudocode

Below is language-agnostic pseudocode illustrating the evolution logic.

```
# Initialize governance state
X          ← X_0                # Initial candidate/team or system state
Lambda_m   ← Lambda_m_init     # Initial multiplicity scalar
P_10       ← [2,3,5,7,11,13,17,19,23,29]

for t in 0 .. T_cycles-1:

    # CSL Gate: Humanity/Dignity/Sovereignty Check
    if not CSL_Gate(X, t):
        log_warning(t, "CSL gate failed - rest/dignity break enforced")
        continue

    # Build prime evolution operator from live type-talk
    Xi_t ← zero_operator()
    for p in P_10:
        U_p ← extract_sectoral_operator(p, t) # From transcripts and tasks
        w_p ← compute_weight(p, t)           # Experts + citizens
        Xi_t ← Xi_t + w_p * U_p
```

```

# Contractivity check (may adapt w_p if too large)
assert norm(Xi_t) <= 1 - epsilon_star

# Measure coherence from outcomes
Gamma_5, Gamma_10, Gamma_20 ← measure_coherence(X, t)

# Update multiplicity scalar
Lambda_m ← update_multiplicity(Lambda_m, Gamma_5, Gamma_10, Gamma_20)

# Apply evolution
X_next ← Xi_t @ X + Lambda_m * T(X)

# Optional: spectral projection
if use_spectral_gates:
    X_next ← sum_over_alpha(E_alpha(t) @ X_next)

# Drift monitoring
drift ← compute_drift(X_next, X, P_10)
if drift > drift_threshold:
    trigger_public_examination(X_next, t)

# Log state and update
log_state(t, norm(X_next), drift, Lambda_m)
X ← X_next

# After T_cycles:
R ← compute_reputation_signature(X, P_10)
return R

```

8.3 Key Supporting Functions

- `CSL_Gate(X, t)`: ensures humane scheduling and dignity constraints.
- `extract_sectoral_operator(p, t)`: uses NLP/LLM tools to map reasoning text into operator approximations.
- `compute_weight(p, t)`: aggregates expert and crowd scores.
- `measure_coherence(X, t)`: compares predicted vs actual outcomes.
- `update_multiplicity`: implements the chosen Λ_m update law.
- `compute_drift`: measures $\delta_{\text{drift}}(t)$.
- `trigger_public_examination`: orchestrates live recall or grilling sessions.

9 Pilot Protocol: Warangal or Equivalent Municipality

9.1 90-Day Pilot Plan

Phase 1 (Days 1–14): Partnership Formation

- Present mathematical specification and governance benefits to municipal leadership.
- Recruit 20–30 candidates for initial Competence Protocol event.

Phase 2 (Days 15–45): Infrastructure Deployment

- Set up live streaming (online and local broadcasters).
- Implement the scoring dashboard for primes and drift metrics.
- Train neutral judges and support staff.

Phase 3 (Days 46–52): Gateway Assessment

- Day 1: Value alignment, orientation, baseline interviews.
- Days 2–5: Deep testing across all 10 pillars, live broadcast.
- Day 6: Long-term consequence cross-examination.
- Day 7: Public scoring, Q&A, initial reputation ledger.

Phase 4 (Days 53–180): Continuous Stream and Evaluation

- Officials govern with ongoing type-talk and recorded decisions.
- Biweekly drift reporting and, if needed, public examinations.
- 6-month coherence analysis vs initial predictions.

9.2 Academic and Open-Source Components

- Academic teams analyze contraction rates and stability vs control municipalities.
- Open-source repositories host code for:
 - reputation calculations,
 - coherence tracking,
 - drift monitoring,
 - streaming dashboards.

10 Prior Art and Defensive Publication Scope

10.1 Claimed Novel Combinations

This document asserts prior art over any system that combines, in the manner described:

1. A prime-indexed set of governance competence pillars, each with associated sectoral operators and weights derived from live reasoning.
2. A Hilbert-space representation of governance state with an evolution of the form

$$X_{t+1} = \Xi(t)X_t + \Lambda_m(t)T(X_t),$$

where $\Xi(t)$ is constructed from public broadcasts and $\Lambda_m(t)$ from long-term outcome coherence.

3. Stability conditions using operator norms and Lipschitz constants ensuring contraction and unique bounded attractors.
4. A spectral drift metric to trigger public cross-examination and recall.
5. A continuously updated reputation ledger expressed as a prime factorization encoding sector strengths.
6. A protocol that ties these mathematical structures to specific practical implementation steps (live type-talk, public Q&A, municipal trials, open-source toolchains).

10.2 Non-Exclusive Use Intention

The intention is to:

- place the described architecture into the public domain as prior art;
- prevent any party from obtaining exclusive rights to the disclosed combination;
- encourage open, cooperative development and deployment of multiplicity-based governance systems.

11 Recommendations and Future Work

11.1 Recommended Next Steps

- Implement a minimal simulation environment to test contraction properties and sensitivity to Λ_m updates.
- Prototype the type-talk parsing pipeline and sectoral operator extraction for synthetic data.
- Conduct a limited-scope pilot (e.g., single council seat) to test live broadcast and scoring with low political risk.
- Expand to multi-seat municipal trials once basic reliability is confirmed.

11.2 Research Directions

- Explore alternative prime sets and pillar decompositions.
- Study robustness to adversarial behavior (e.g., candidates gaming type-talk patterns).
- Investigate links between spectral properties of $\Xi(t)$ and classical political theory concepts (e.g., legitimacy, authority types).
- Extend the model to multi-agent, multi-jurisdictional governance (federal and global scales).

Conclusion

We have formalized a governance architecture in which leaders, policies, and even the rules of the system evolve under a mathematically explicit, prime-indexed evolution operator. By integrating Multiplicity Theory with a concrete, broadcast-based assessment protocol, this framework renders governance into an observable, contractive process with built-in feedback loops and stability analysis. This document provides a complete specification—conceptual, mathematical, algorithmic, and procedural—sufficient to serve as defensive prior art and to guide real-world implementation and evaluation.

Mathematical Appendix

A Preliminaries and Notation

Let $(\mathcal{H}, \langle \cdot, \cdot \rangle)$ be a real or complex Hilbert space with induced norm

$$\|X\| = \sqrt{\langle X, X \rangle}, \quad X \in \mathcal{H}.$$

For any bounded linear operator $A : \mathcal{H} \rightarrow \mathcal{H}$, its operator norm is

$$\|A\| = \sup_{\|X\|=1} \|AX\|.$$

We denote by $\mathcal{B}(\mathcal{H})$ the Banach space of bounded operators on $\mathcal{H}$ with norm $\|\cdot\|$. Symbols and definitions used throughout:

- $P_{10} = \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29\}$: the set of 10 prime indices for competence pillars.
- For each $p \in P_{10}$ and discrete time $t \in \mathbb{N}$, $U_p(t) \in \mathcal{B}(\mathcal{H})$ is the sectoral operator associated to prime p at time t .
- $w_p(t) \in \mathbb{R}$ is the weight assigned to sector p at time t based on real-time scoring and feedback.
- The evolution operator $\Xi(t) \in \mathcal{B}(\mathcal{H})$ is

$$\Xi(t) = \sum_{p \in P_{10}} w_p(t) U_p(t).$$

- $\Lambda_m(t) \in \mathbb{R}$ is the multiplicity scalar at time t , updated from outcome coherence.
- $T : \mathcal{H} \rightarrow \mathcal{H}$ is a (generally nonlinear) transform with Lipschitz constant $L_T > 0$:

$$\|T(X) - T(Y)\| \leq L_T \|X - Y\|, \quad \forall X, Y \in \mathcal{H}.$$

- $c = \sup_t \|\Lambda_m(t)\| L_T$ is the *multiplicity coupling*.

The discrete-time governance evolution law is:

$$X_{t+1} = \Xi(t)X_t + \Lambda_m(t)T(X_t), \quad t = 0, 1, 2, \dots \quad (1)$$

We assume throughout that the time dependence is such that all quantities are well-defined and uniformly bounded across t .

B Operator Norm Bounds for $\Xi(t)$

B.1 Construction and Basic Bounds

Recall

$$\Xi(t) = \sum_{p \in P_{10}} w_p(t) U_p(t).$$

We impose the following structural conditions:

1. For each $p \in P_{10}$ and t , $U_p(t)$ is bounded with

$$\|U_p(t)\| \leq M_p$$

for some constants $M_p \geq 0$, independent of t .

2. The weights $w_p(t)$ are bounded uniformly in time:

$$|w_p(t)| \leq \omega_p, \quad \forall t.$$

[Norm Bound for $\Xi(t)$] Under the assumptions above,

$$\|\Xi(t)\| \leq \sum_{p \in P_{10}} |w_p(t)| \|U_p(t)\| \leq \sum_{p \in P_{10}} \omega_p M_p.$$

In particular, if

$$\sum_{p \in P_{10}} \omega_p M_p \leq 1 - \varepsilon_\star$$

for some $\varepsilon_\star > 0$, then

$$\|\Xi(t)\| \leq 1 - \varepsilon_\star \quad \forall t.$$

Proof. For any $X \in \mathcal{H}$, we have:

$$\|\Xi(t)X\| = \left\| \sum_{p \in P_{10}} w_p(t) U_p(t) X \right\| \leq \sum_{p \in P_{10}} |w_p(t)| \|U_p(t)X\| \leq \sum_{p \in P_{10}} |w_p(t)| \|U_p(t)\| \|X\|.$$

Taking the supremum over $\|X\| = 1$ yields:

$$\|\Xi(t)\| \leq \sum_{p \in P_{10}} |w_p(t)| \|U_p(t)\|.$$

Using the uniform bounds $|w_p(t)| \leq \omega_p$ and $\|U_p(t)\| \leq M_p$ gives:

$$\|\Xi(t)\| \leq \sum_{p \in P_{10}} \omega_p M_p.$$

If the sum $\sum_{p \in P_{10}} \omega_p M_p$ is bounded by $1 - \varepsilon_\star$, the final statement follows immediately. $\square$

B.2 Design Implications

In practice, this lemma provides a design constraint:

- either the weights $w_p(t)$ must be adaptively normalized,
- or the underlying operators $U_p(t)$ must be scaled,

so that the aggregate norm of $\Xi(t)$ remains uniformly bounded by $1 - \varepsilon_\star$. This ensures the linear part of the evolution remains contractive.

C Lipschitz Bound and Multiplicity Coupling

[Lipschitz Continuity of T] We say $T : \mathcal{H} \rightarrow \mathcal{H}$ is Lipschitz continuous if there exists $L_T > 0$ such that

$$\|T(X) - T(Y)\| \leq L_T \|X - Y\|, \quad \forall X, Y \in \mathcal{H}.$$

This bound is natural when T aggregates prediction–outcome discrepancies: small changes in the underlying state should not lead to arbitrarily large changes in feedback.

[Multiplicity Coupling] For $\Lambda_m(t) \in \mathbb{R}$, define the multiplicity coupling constant

$$c := \sup_t \|\Lambda_m(t)\| L_T.$$

By design of the update rule for $\Lambda_m(t)$, we can enforce c to be finite and, in fact, to satisfy a strict inequality relative to the contraction margin.

D Proof of the Stability and Contraction Theorem

We now provide an explicit proof of the discrete-time stability theorem for the evolution law (1).

D.1 Setup

Let X_t, Y_t be two trajectories in $\mathcal{H}$ evolving under the same operators $\Xi(t)$, T , and scalars $\Lambda_m(t)$, but with possibly different initial states X_0, Y_0 . We assume:

1. $\|\Xi(t)\| \leq 1 - \varepsilon$ for some $\varepsilon > 0$ and all t .
2. T is Lipschitz with constant L_T .
3. $c = \sup_t |\Lambda_m(t)| L_T < \varepsilon$.

D.2 Main Inequality

From (1), we have:

$$X_{t+1} - Y_{t+1} = \Xi(t)X_t + \Lambda_m(t)T(X_t) - (\Xi(t)Y_t + \Lambda_m(t)T(Y_t)) = \Xi(t)(X_t - Y_t) + \Lambda_m(t)(T(X_t) - T(Y_t)).$$

Taking norms:

$$\|X_{t+1} - Y_{t+1}\| \leq \|\Xi(t)\| \|X_t - Y_t\| + |\Lambda_m(t)| \|T(X_t) - T(Y_t)\|.$$

Using the Lipschitz bound on T , we get:

$$\|T(X_t) - T(Y_t)\| \leq L_T \|X_t - Y_t\|.$$

Thus:

$$\|X_{t+1} - Y_{t+1}\| \leq \|\Xi(t)\| \|X_t - Y_t\| + |\Lambda_m(t)| L_T \|X_t - Y_t\| = (\|\Xi(t)\| + |\Lambda_m(t)| L_T) \|X_t - Y_t\|.$$

Using the uniform bounds:

$$\|\Xi(t)\| \leq 1 - \varepsilon, \quad |\Lambda_m(t)| L_T \leq c,$$

we obtain:

$$\|X_{t+1} - Y_{t+1}\| \leq (1 - \varepsilon + c) \|X_t - Y_t\|.$$

D.3 Iterated Bound

Iterating the inequality:

$$\|X_{t+1} - Y_{t+1}\| \leq (1 - \varepsilon + c)\|X_t - Y_t\| \leq (1 - \varepsilon + c)^2\|X_{t-1} - Y_{t-1}\| \leq \dots$$

we obtain by induction:

$$\|X_t - Y_t\| \leq (1 - \varepsilon + c)^t\|X_0 - Y_0\|.$$

Since $c < \varepsilon$, we have $1 - \varepsilon + c < 1$. Let

$$q := 1 - \varepsilon + c \in (0, 1).$$

Then:

$$\|X_t - Y_t\| \leq q^t\|X_0 - Y_0\|.$$

Because $q^t \rightarrow 0$ as $t \rightarrow \infty$, we conclude:

$$\|X_t - Y_t\| \rightarrow 0 \quad \text{as } t \rightarrow \infty.$$

D.4 Existence of a Unique Bounded Attractor

The inequality above implies that for any two trajectories under the same sequence of operators and scalars, the distance between them decays to zero. In particular:

- For fixed operator sequence $\{\Xi(t)\}$, nonlinearity T , and scalar sequence $\{\Lambda_m(t)\}$ satisfying the stated bounds, there exists a unique limit trajectory (up to bounded perturbations).
- Any initial state X_0 converges toward the same attractor (a bounded subset—possibly a single point, fixed cycle, or small compact set depending on additional structure).

If, in addition, the system is autonomous or asymptotically autonomous (e.g., $\Xi(t) \rightarrow \Xi_\infty$, $\Lambda_m(t) \rightarrow \Lambda_\infty$), one can show that the attractor is a fixed point or invariant set of the limiting dynamics.

Remark (Governance Interpretation). In governance terms, this theorem states that under sustained application of the protocol:

- different initial political states (legacy regimes, initial candidate sets) eventually converge to the same competence-dominated regime;
- the system is resilient to initial conditions and transient perturbations, provided the contractivity and coupling conditions are maintained.

E Bounds for the Coherence-Driven Update $\Lambda_m(t)$

E.1 Coherence Aggregation

Let $\{\Gamma(D, t)\}_D$ denote the coherence values for decisions D evaluated at time t , with each $\Gamma(D, t) \geq 0$. Define an aggregated coherence:

$$\bar{\Gamma}(t) = \sum_D \beta_D \Gamma(D, t),$$

where $\beta_D \geq 0$ and $\sum_D \beta_D \leq 1$ (or more generally bounded). Suppose each $\Gamma(D, t)$ satisfies

$$0 \leq \Gamma(D, t) \leq G_{\max},$$

then

$$0 \leq \bar{\Gamma}(t) \leq G_{\max} \sum_D \beta_D \leq G_{\max}.$$

E.2 Example Update Rule

Consider the exponential decay update:

$$\Lambda_m(t+1) = \Lambda_m(t) \exp(-\alpha \bar{\Gamma}(t)),$$

with $\alpha > 0$.

Assuming $\Lambda_m(0)$ is bounded, we have:

$$|\Lambda_m(t+1)| = |\Lambda_m(t)| \exp(-\alpha \bar{\Gamma}(t)) \leq |\Lambda_m(t)| \leq |\Lambda_m(0)|,$$

since $\exp(-\alpha \bar{\Gamma}(t)) \leq 1$. Thus:

$$|\Lambda_m(t)| \leq |\Lambda_m(0)|, \quad \forall t.$$

Consequently,

$$c = \sup_t |\Lambda_m(t)|_{L_T} \leq |\Lambda_m(0)|_{L_T}.$$

To satisfy $c < \varepsilon$, it suffices to choose

$$|\Lambda_m(0)| < \frac{\varepsilon}{L_T}.$$

E.3 Tighter Bounds with Lower Coherence

If $\bar{\Gamma}(t)$ is bounded below by a positive constant $\gamma_0 > 0$ (e.g., because some error is always present), then

$$|\Lambda_m(t+1)| = |\Lambda_m(t)| \exp(-\alpha \bar{\Gamma}(t)) \leq |\Lambda_m(t)| \exp(-\alpha \gamma_0).$$

Iterating:

$$|\Lambda_m(t)| \leq |\Lambda_m(0)| \exp(-\alpha \gamma_0 t).$$

Thus $|\Lambda_m(t)| \rightarrow 0$ exponentially, making the coupling c not only bounded but decaying over time, further strengthening convergence.

F Spectral Drift and Trigger Conditions

F.1 Definition and Bounds

Let $\Phi(p, t) \in \mathbb{R}$ denote a phase-like scalar observable (e.g., an angle or normalized score) for sector p at time t . Define:

$$\delta_{\text{drift}}(t) = \sum_{p \in P_{10}} \kappa_p [\Phi(p, t) - \Phi(p, t-1)],$$

where $\kappa_p \in \mathbb{R}$ encode importance or sensitivity per sector.

Assume:

- each $\Phi(p, t)$ is bounded; say $|\Phi(p, t)| \leq B_p$,
- and variation per step is bounded: $|\Phi(p, t) - \Phi(p, t-1)| \leq \Delta_p$.

Then:

$$|\delta_{\text{drift}}(t)| \leq \sum_{p \in P_{10}} |\kappa_p| |\Phi(p, t) - \Phi(p, t-1)| \leq \sum_{p \in P_{10}} |\kappa_p| \Delta_p.$$

F.2 Trigger Threshold

Choose a drift threshold $\Theta > 0$ such that:

$$\Theta < \sum_{p \in P_{10}} |\kappa_p| \Delta_p.$$

When $|\delta_{\text{drift}}(t)| > \Theta$, the system triggers cross-examination or recall:

- The threshold Θ can be set as a fraction of the theoretical maximum drift,
- or adaptively based on historical drift distributions.

Though δ_{drift} itself does not directly enter the contraction proof, it serves as a diagnostic that can induce human or procedural interventions (e.g., resetting certain components of X_t , or modifying $U_p(t)$ or $w_p(t)$) when the system deviates from stable behavior patterns.

G Spectral Decomposition and Projection

G.1 Spectral Projectors

Suppose for each t we have a family of orthogonal projectors $\{E_\alpha(t)\}_\alpha$ in $\mathcal{B}(\mathcal{H})$ such that:

$$E_\alpha(t)^2 = E_\alpha(t), \quad E_\alpha(t)^* = E_\alpha(t), \quad E_\alpha(t)E_\beta(t) = 0 \text{ for } \alpha \neq \beta,$$

and

$$I = \sum_{\alpha} E_\alpha(t).$$

We may interpret each $E_\alpha(t)$ as projecting onto a spectral band of acceptable governance states (e.g., respecting constitutional constraints).

G.2 Projected Evolution

Instead of using

$$X_{t+1} = \Xi(t)X_t + \Lambda_m(t)T(X_t),$$

we apply a projection:

$$X_{t+1} = \sum_{\alpha} E_\alpha(t) (\Xi(t)X_t + \Lambda_m(t)T(X_t)).$$

[Norm Preservation Under Orthogonal Projection] For any orthogonal projector P , $\|PX\| \leq \|X\|$ for all $X \in \mathcal{H}$.

Proof. Because P is an orthogonal projector, $P = P^* = P^2$ and the range of P is a closed subspace. The operator norm of an orthogonal projector satisfies $\|P\| = 1$, hence:

$$\|PX\| \leq \|P\| \|X\| = \|X\|.$$

□

Applying $\sum_{\alpha} E_{\alpha}(t)$ (with orthogonal projectors) cannot increase the norm of the state. Thus, if the unprojected evolution is contractive, so is the projected one (or more so).

H Continuous-Time Evolution and Boundedness

H.1 Continuous-Time Model

We consider:

$$\dot{X}(\tau) = F(\tau)X(\tau) + \Lambda_m(\tau)T(X(\tau)),$$

with $F(\tau) \in \mathcal{B}(\mathcal{H})$ for each τ .

Assume:

- $\|F(\tau)\| \leq -\delta < 0$ in a suitable dissipative sense (e.g., $\langle F(\tau)X, X \rangle \leq -\delta\|X\|^2$) or that $F(\tau)$ generates a contraction semigroup,
- T and $\Lambda_m(\tau)$ satisfy Lipschitz and boundedness conditions similar to the discrete case.

H.2 Energy Estimate

A standard approach (via Grönwall-type inequalities) gives, under suitable assumptions:

$$\frac{d}{d\tau}\|X(\tau)\| \leq -\delta\|X(\tau)\| + |\Lambda_m(\tau)|L_T\|X(\tau)\|.$$

Thus:

$$\frac{d}{d\tau}\|X(\tau)\| \leq (-\delta + |\Lambda_m(\tau)|L_T)\|X(\tau)\|.$$

If, for all τ ,

$$|\Lambda_m(\tau)|L_T \leq \delta - \epsilon$$

for some $\epsilon > 0$, then:

$$\frac{d}{d\tau}\|X(\tau)\| \leq -\epsilon\|X(\tau)\|.$$

Integrating yields:

$$\|X(\tau)\| \leq \|X(0)\|e^{-\epsilon\tau},$$

demonstrating exponential decay and boundedness. The continuous-time argument parallels the discrete-time contraction, with differential inequalities replacing recurrence relations.

I Summary of Conditions and Guarantees

For clarity, we summarize the key inequalities and their roles:

- $\|\Xi(t)\| \leq 1 - \varepsilon$: ensures linear part is contractive.
- $\|T(X) - T(Y)\| \leq L_T\|X - Y\|$: ensures feedback map is well-behaved.
- $c = \sup_t |\Lambda_m(t)|L_T < \varepsilon$: keeps nonlinear coupling below contraction margin.
- $|\delta_{\text{drift}}(t)| \leq \sum_{p \in P_{10}} |\kappa_p| \Delta_p$: bounds spectral drift; thresholding this triggers interventions.
- $|\Lambda_m(t)| \leq |\Lambda_m(0)|$ (and possibly exponential decay): ensures multiplicity scalar does not blow up.

Under these conditions, the prime-indexed recursive governance system:

1. admits a unique bounded attractor (modulo structural details),
2. contracts trajectories from diverse initial conditions,
3. allows systematic detection of anomalous behavior via drift,
4. can accommodate further constraints via spectral projections without compromising stability.

This completes the explicit mathematical underpinning for the governance evolution operator and its stability properties.

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